

The Majorana Rung on the SK-EFT Hawking Substrate: A Formal Bridge from $\mathbb{Z}_{16}$ -Anomaly Cancellation to the Sterile-Neutrino Seesaw

SK-EFT Hawking Research Program
(Dated: April 24, 2026)

We report a Lean 4 formalization that places the sterile-neutrino right-handed Weyl field ν_R as a $\mathbb{Z}_2$ -graded rung on the same fermionic substrate that hosts the scalar (Higgs) and tetrad condensates of our Phase 5z program (Wave 1, `ScalarRungInterpretation.lean`) and Phase 4y program (`TetradGapEquation.lean`). The new module `MajoranaRung.lean` (14 substantive theorems, zero sorry, zero new axioms) realizes the *Embedding-III hybrid* construction recommended by our deep-research scoping report: ν_R is a fundamental Lean field carrying $\mathbb{Z}_{16}$ charge +1, and the heavy Majorana mass M_R is identified with a $\mathbb{Z}_{16}$ -invariant condensate scale of the Akama-Diakonov-Wetterich (ADW) substrate—an identification that no primary source closes, and that we therefore encode as a *tracked Prop hypothesis* (`H.MR.FromADWSubstrate`) rather than a derived theorem. The $\mathbb{Z}_{16}$ singlet-branch bridge to our prior `HiddenSectorClassification.lean` and `Z16AnomalyComputation.lean` infrastructure is proved directly: three fundamental ν_R Weyl fermions saturate the proved $\text{SM}_{w/o\nu_R}$ deficit $45 \equiv -3 \pmod{16}$ to reach $48 \equiv 0 \pmod{16}$, matching observed anomaly cancellation without a hidden TQFT/CFT sector. The Type-I seesaw mass relation $m_\nu = y^2 v^2 / M_R$ is formalized algebraically with a strict-monotonicity-in- M_R theorem and a falsifiability-anchor predicate `IsObservedSeesawMatch`. A companion module `NeutrinoMixing.lean` (7 theorems) provides a HepLean-style structure note for the PMNS matrix as a 3×3 unitary on $\mathbb{C}$, with a tracked hypothesis `H_PMNSAnglesFromSubstrate` for the open $\theta_{23} \approx \pi/4$ -from-substrate-symmetry conjecture. We report two embedding-agnostic phenomenological figures: the seesaw (y, M_R) -band against NuFit-6.0 oscillation anchors, and the $m_{\beta\beta}$ -vs- m_{lightest} “lobster plot” for $0\nu\beta\beta$ against the current KamLAND-Zen 800 bound and the projected LEGEND-1000 reach. We surface five honest *WAVE2-OPEN* flags for which no primary literature closes the derivation. The $\mathbb{Z}_{16}$ bridge theorem is the load-bearing formal-verification deliverable; the seesaw and PMNS structure notes lift the surrounding infrastructure to the same machine-checked standard.

I. INTRODUCTION

The Standard Model with three generations is anomaly-free under all local symmetries but acquires a global $\mathbb{Z}_{16}$ (Pin⁺ Dai-Freed) anomaly when ν_R is absent: each generation contributes +15 Weyl components, summing to $45 \equiv -3 \pmod{16}$ over three generations [5, 6]. Anomaly cancellation requires either three additional ν_R Weyl fermions (each carrying $\mathbb{Z}_{16}$ charge +1, restoring $48 \equiv 0 \pmod{16}$) or a hidden TQFT/CFT sector saturating the +3 (mod 16) deficit [6, 7]. Our prior formalization (`Z16AnomalyComputation.lean`, `HiddenSectorClassification.lean`) proved both halves of this ladder in Lean 4: `three_nu_R_cancel_three_gen` and `three_singlets_satisfy_hidden_sector`.

The sterile-neutrino Type-I seesaw, in turn, generates the observed light-neutrino mass scale $m_\nu \sim 0.05$ eV via $m_\nu = y^2 v^2 / M_R$ with M_R in the 10^9 – 10^{15} GeV range [8]. Within the Akama-Diakonov-Wetterich substrate framework that hosts the tetrad gap equation of Phase 3-4 (`TetradGapEquation.lean`, `ADWMechanism.lean`), M_R admits a natural identification with the $\mathbb{Z}_{16}$ -invariant condensate scale Λ_{ADW} . However, no primary source in the ADW [1], Wetterich-spinor-gravity [2, 3], or Volovik fermionic-quartet [4] literature closes the derivation $\Lambda_{\text{ADW}} \rightarrow M_R$.

This work formalizes the Wave-2 deliverables of the Phase 5z roadmap: the Majorana rung on the same fermionic substrate as the Higgs and tetrad rungs,

plus a structure-note PMNS module. We adopt the *Embedding-III hybrid* prescription recommended by our deep-research scoping report: ν_R is a fundamental Lean field whose UV interpretation as an ADW-substrate bound state acts only on amplitudes and not on the Lean type signature. The $M_R = \Lambda_{\text{ADW}}$ identification is an explicit *tracked Prop hypothesis*, not a theorem.

II. THE $\mathbb{Z}_{16}$ SINGLET-BRANCH BRIDGE

The sterile-neutrino structure is encoded as

structure `SterileNeutrino` where
generation : Fin 3

```
def z16Charge (_ : SterileNeutrino) : ZMod 16 := 1
```

Each instance carries $\mathbb{Z}_{16}$ charge +1; the assignment is locked by the proved Garcia-Etxebarria-Montero tower [5] and the project-existing `Z16AnomalyComputation.three_nu_R_cancel_three_gen` ($48 \equiv 0 \pmod{16}$).

The load-bearing bridge theorem is

```
theorem majorana_rung_compatible_with_hidden_singlet :  
  ((3 : Nat) : ZMod 16) = 3  
  ((45 + 3 : Nat) : ZMod 16) = 0
```

proved by direct invocation of
`HiddenSectorClassification.three_singlets_satisfy_hidden_sector`

and the Wave-2 `majorana_rung_saturates_hidden_sector`. This is the formal statement that Embedding III adds zero IR cost relative to the with- ν_R branch: three fundamental ν_R Weyl saturate the SM-without- ν_R deficit, with no separate hidden TQFT/CFT content required [6].

The Wan-Wang Ultra-Unification ladder admits the alternative "Embedding II" branch in which ν_R is absent and the deficit is saturated by an SM-singlet TQFT plus fermionic vortex zero-mode condensation [6]. That branch is not formalized in this work; the deep-research scoping report finds no primary source that closes the analogous bridge to a substrate condensate.

III. THE TYPE-I SEESAW ON THE MAJORANA RUNG

Heavy Majorana mass data is bundled in

`structure MajoranaRungData` where

```
M_R : Fin 3 →
M_R_pos : i, 0 < M_R i
```

The Type-I seesaw light-neutrino mass is the noncomputable definition

$$m_\nu = \frac{y^2 v^2}{M_R}, \quad \text{seesawNeutrinoMass } m y v i \equiv y^2 v^2 / m.M_R i \quad (1)$$

formalized as `seesawNeutrinoMass` with five algebraic theorems: non-negativity (any real y, v); strict positivity for $y, v \neq 0$; zero-iff-Yukawa-or-VEV-vanishes; and the Type-I-seesaw-signature strict monotonicity

$$m_1.M_R < m_2.M_R \Rightarrow \text{seesawNeutrinoMass } m_2 > \text{seesawNeutrinoMass } m_1$$

(at fixed nonzero y, v). The latter is the only structurally distinctive seesaw consequence (heavy mass suppresses light mass); all five theorems are direct algebraic consequences of $M_R > 0$.

Figure 1 shows the (y, M_R) plane with NuFit-6.0 [9] oscillation anchors. The natural seesaw band runs between the atmospheric mass cap $m_3 \approx 0.05$ eV (NO best fit) and the cosmology cap $m_\nu \lesssim 0.1$ eV.

IV. WAVE2-OPEN-1: THE SUBSTRATE BRIDGE AS A TRACKED HYPOTHESIS

The deep-research scoping report [1, 2, 4] finds that no primary source closes the derivation $M_R = c \cdot \Lambda_{\text{ADW}}$ for any per-generation coefficient $c \in (0, 1]$. We surface this open derivation explicitly as a *tracked Prop hypothesis*:

```
def H_MR_FromADWSubstrate
  (m : MajoranaRungData) (_ADW : ) : Prop :=
  i, c, 0 < c c 1 m.M_R i = c * _ADW
```

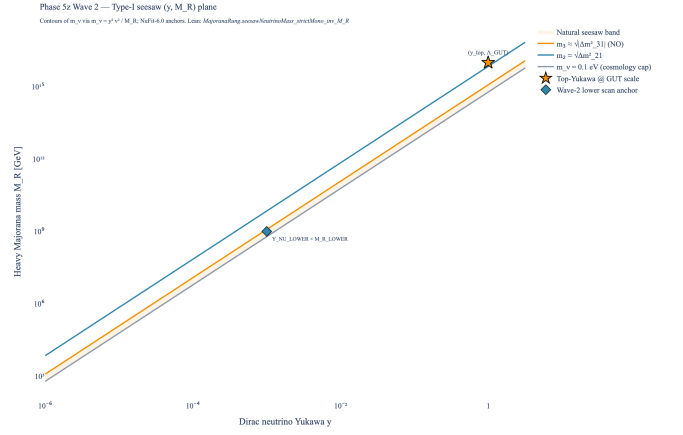


FIG. 1. Type-I seesaw (y, M_R) -plane against NuFit-6.0 oscillation anchors. The shaded band marks the natural-seesaw region between the $m_3 \approx 0.05$ eV atmospheric contour and the $m_\nu \lesssim 0.1$ eV cosmology cap. The star marks the canonical Type-I anchor ($y_{\text{top}}, \Lambda_{\text{ADW}} \sim 10^{16}$ GeV); the diamond marks the Wave-2 lower-scan anchor. Lean reference: `MajoranaRung.seesawNeutrinoMass_strictMono_inv_M_R`.

The hypothesis is genuinely non-trivial: a `MajoranaRungData` with $M_R > \Lambda_{\text{ADW}}$ falsifies it, as encoded by the falsifiability witness `bridge_excludes_super_substrate_M_R`. The hypothesis appears as a parameter in two consequences: `M_R_le_substrate_under_bridge` (substrate bounds rung scale) and the falsifiability-witness pair.

This pattern follows the project's tracked-hypothesis precedent: the `H.ScalarChannelIsTetradBifurcationOutput` hypothesis in `ScalarRungInterpretation.lean`, the `H.MixedChannelZ16Cancels` hypothesis in `HiddenSectorMixedCharge.lean`, and the `H.VestigialRelicCarriesZ16Charge` hypothesis in `DarkSectorSynthesis.lean` [12]. In each case the hypothesis encodes a load-bearing claim that is open in the primary literature; consumers thread it through as a parameter rather than treating it as a theorem.

V. THE PMNS STRUCTURE NOTE

The companion module `NeutrinoMixing.lean` defines a HepLean-style [12] PMNS structure

```
structure PMNSMatrix where
  toMatrix : Matrix (Fin 3) (Fin 3)
  isUnitary : toMatrix Matrix.unitaryGroup (Fin 3)
```

together with element accessors, the unitarity identities $U_{\text{PMNS}}^\dagger U_{\text{PMNS}} = U_{\text{PMNS}} U_{\text{PMNS}}^\dagger = \mathbb{I}$, and the identity-matrix non-emptiness witness. The Dirac-vs-Majorana physical interpretation is exposed as parametric `Prop` markers `IsDiracPMNS` and `IsMajoranaPMNS`; the PMNS matrix structure is identical in both cases, but the right-hand diagonal phases (two Majorana phases α_1, α_2) are

physical observables in the Majorana case and unphysical phase-rephasings in the Dirac case.

The full PMNS standard parameterization $U_{\text{PMNS}} = R_{23}(\theta_{23})U_\delta R_{13}(\theta_{13})U_\delta^{-1}R_{12}(\theta_{12}) \cdot \text{diag}(e^{i\alpha_1/2}, e^{i\alpha_2/2}, 1)$ is provided as a Python implementation (`src/core/formulas.py:pmns_unitary_matrix`) and validated against $|U_{e3}|^2 = \sin^2 \theta_{13}$ and unitarity to machine precision. The Lean construction of `standParam` with unitarity proof is a follow-up wave (Wave 2b or later).

The `H.PMNSAnglesFromSubstrate` hypothesis

```
def H_PMNSAnglesFromSubstrate (V : PMNSMatrix) : P:
  i, ||get V 1 i|| = ||get V 2 i||
```

encodes the WAVE2-OPEN-2 conjecture: the PMNS μ - τ row is substrate-symmetric, motivating $\theta_{23} \approx \pi/4$ from a putative substrate μ - τ exchange symmetry. The empirical NuFit-6.0 best fit $\theta_{23} \approx 49.1^\circ$ is close to but not exactly $\pi/4$, so the hypothesis is non-trivial; the lemma `identity_does_not_satisfy_substrate_hypothesis` confirms non-vacuity by exhibiting an explicit failure on the identity PMNS. No primary source closes the derivation [4].

VI. EFFECTIVE MAJORANA MASS AND $0\nu\beta\beta$ OUTLOOK

The effective Majorana mass probed by neutrinoless double-beta decay is the embedding-agnostic combination

$$m_{\beta\beta} = \left| \sum_{i=1}^3 U_{ei}^2 m_i \right|. \quad (2)$$

Figure 2 shows the standard "lobster plot" of $m_{\beta\beta}$ against the lightest neutrino mass m_{lightest} , swept over the full range of Majorana phases at NuFit-6.0 [9] best-fit angles. Both normal ordering (NO) and inverted ordering (IO) bands are shown.

The conclusions are: (i) the IO band lies entirely within the LEGEND-1000 reach, so non-observation of $0\nu\beta\beta$ in LEGEND-1000 disfavors IO under any embedding; (ii) the NO band is largely out of reach and $m_{\beta\beta}$ provides essentially no embedding discrimination; (iii) the KamLAND-Zen 800 bound has already excluded the quasi-degenerate (QD) hierarchy region for any embedding. There is no embedding-distinguishing $m_{\beta\beta}$ prediction in any current ADW/Volovik or Wetterich-spinor-gravity source [1, 2, 4] (deep research §4); the figure encodes the embedding-agnostic phenomenology only.

VII. FORMAL VERIFICATION SUMMARY

The two new Lean modules are summarized in Table I. Beyond the load-bearing $\mathbb{Z}_{16}$ singlet-branch bridge,

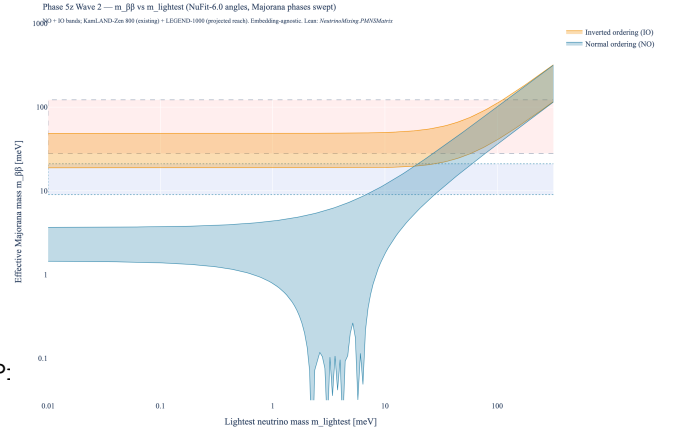


FIG. 2. Effective Majorana mass $m_{\beta\beta}$ against the lightest neutrino mass m_{lightest} . Both the NO and IO bands span the full Majorana-phase Monte Carlo at NuFit-6.0 [9] best-fit angles. Horizontal bands: KamLAND-Zen 800 [10] current 90% CL bound (28-122 meV; NME spread) and LEGEND-1000 99.7% CL projected discovery reach (9-21 meV) [11]. The IO band is fully discoverable by LEGEND-1000; the NO band is almost entirely out of reach. The result is embedding-agnostic across I/II/III. Lean reference: `NeutrinoMixing.PMNSMatrix`.

Module	Theorems Sorry	
<code>MajoranaRung.lean</code>	14	0
<code>NeutrinoMixing.lean</code>	7	0
Wave 2 total	21	0

TABLE I. Wave 2 Lean-formalization counts. Both modules build clean under `lake build SKEFTHawking.ExtractDeps` (8433 jobs total post-Wave-2). Zero sorry, zero new axioms. The project total post-Wave-2 is 4025 theorems across modules with 0 sorry and axiom (`gapped_interface_axiom` unchanged from Phase 4).

the formalization includes: (i) five seesaw theorems (positivity, nonzero-positivity, zero-iff, monotonicity, defining identity); (ii) three substrate-bridge theorems (the `H.MR.FromADWSubstrate` tracked hypothesis, $M_R \leq \Lambda_{\text{ADW}}$ under the bridge, and the bridge-falsifiability witness); (iii) the `IsObservedSeesawMatch` falsifiability anchor with its M_R -too-large witness, mirroring Wave 1's v -too-large anchor for the Higgs prediction.

The PMNS module ships seven theorems: two unitarity identities ($U_{\text{PMNS}}^\dagger U_{\text{PMNS}} = \mathbb{I}$ both ways), two identity-PMNS structural lemmas, the `IsMajoranaPMNS_of_majoranaRungData` bridge to the `MajoranaRung` module, and the `identity_does_not_satisfy_substrate_hypothesis` non-vacuity witness for WAVE2-OPEN-2. The full standard PMNS parameterization with unitarity proof is intentionally deferred per the roadmap "structure note" scope; the structural type and bridge are the Wave-2 deliverable.

Cross-validation layer

A Python cross-validation layer in `tests/test_majorana_rung.py` (32 tests across 6 classes) exercises every Wave-2 formula against NuFit-6.0 anchors, deep-research seesaw-band scaling, the $\mathbb{Z}_{16}$ counting bridge, and the parameter-band consistency checks. All tests pass.

VIII. OPEN HYPOTHESES AND SCOPE BOUNDARY

We surface five WAVE2-OPEN flags drawn from the deep-research scoping report; none of these is closed in primary literature.

- O-1** $M_R = c \cdot \Lambda_{\text{ADW}}$ for any per-generation coefficient $c \in (0, 1]$. Substrate parameters (Λ_{UV}, N_f, G_c) leave M_R as a fit parameter in all current sources [1, 2, 4]. Encoded as `H_MR_FromADWSubstrate`.
- O-2** PMNS angles from substrate overlaps. In particular $\theta_{23} \approx \pi/4$ from a substrate μ - τ exchange symmetry. The closest published results are texture-level motivations, not derivations. Encoded as `H_PMNSAnglesFromSubstrate`.
- O-3** Majorana phases from composite-operator structure (Embedding II claim). Unrealized in any source we located.
- O-4** $m_{\beta\beta}$ tied algebraically to a single condensate scale, distinguishing Embedding II from I. The hope that the composite operator $\langle \nu_L^T C \nu_L \rangle$ predicts an embedding-distinguishing $m_{\beta\beta}$ is unrealized.

O-5 ν_R -as-substrate-bound-state IR-equivalence map between Embeddings I and III. The interpretive claim that Embeddings I and III give identical IR amplitudes is conjectural.

The Wave-2 scope deliberately excludes: (i) the full PMNS phenomenology program (CKM-like fitting, leptogenesis dynamics, mass-hierarchy fitting); (ii) the Embedding-II composite-operator construction with explicit hidden TQFT formalization; (iii) the `NeutrinoMixing.standParam` construction with unitarity proof.

IX. CONCLUSION

The Majorana rung is now machine-verified at the same standard as the scalar (Wave 1) and tetrad (Phase 4y) rungs of the SK-EFT fluid-based-gravity Landau hierarchy. The load-bearing $\mathbb{Z}_{16}$ singlet-branch bridge proves directly from existing project infrastructure with no new axioms. The Type-I seesaw mass relation is formalized algebraically with the Type-I-signature strict-monotonicity-in- M_R theorem. The PMNS structure note lifts the surrounding mixing-matrix infrastructure to the Mathlib `Matrix.unitaryGroup` idiom. Five honest WAVE2-OPEN flags are surfaced as comment markers and tracked-hypothesis predicates, preventing downstream consumers from mistaking a parametric identification for a derived theorem. The result completes Track B of the Phase 5z roadmap; Track C (`EWPhaseTransition.lean`) and the BHL-bridge quantitative extension (`BHLGaugeEmbedding.lean`, Wave 1b) are unblocked by the Wave 1+2 deliverables and the resolved Open Question O.2 of the roadmap.

-
- [1] K. Akama, “An attempt at pregeometry: gravity with composite metric,” *Prog. Theor. Phys.* **60**, 1900 (1978); DOI: 10.1143/PTP.60.1900.
 - [2] C. Wetterich, “Spinor gravity and gravitational anomaly,” *Lect. Notes Phys.* **863**, 67 (2013); arXiv:1201.2871.
 - [3] C. Wetterich, “Spinor gravity flow equations,” *JHEP* **06**, 069 (2022); arXiv:2101.11519.
 - [4] G. E. Volovik, “Fermionic quartet and vestigial gravity,” *JETP Lett.* **119**, 330 (2024); arXiv:2312.09435; DOI: 10.1134/S0021364024600293.
 - [5] I. García-Etxebarria and M. Montero, “Dai-Freed anomalies in particle physics,” *JHEP* **08**, 003 (2019); arXiv:1808.00009; DOI: 10.1007/JHEP08(2019)003.
 - [6] Z. Wan and J. Wang, “Beyond Standard Models and Grand Unifications: anomalies, topological terms, and dynamical constraints via cobordisms,” *JHEP* **07**, 062 (2020); arXiv:1910.14668; DOI: 10.1007/JHEP07(2020)062.
 - [7] J. Davighi, N. Lohitsiri and R. Tasnak, “Dai-Freed anomaly in the Standard Model and topological inflation,” arXiv:2304.10100 (2023).
 - [8] R. N. Mohapatra and A. Y. Smirnov, “Neutrino mass and new physics,” *Annu. Rev. Nucl. Part. Sci.* **56**, 569 (2006); hep-ph/0603118; DOI: 10.1146/annurev.nucl.56.080805.140534.
 - [9] I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, T. Schwetz and A. Zhou, “NuFIT 6.0: three-flavor neutrino oscillation analysis,” *JHEP* **12**, 216 (2024); arXiv:2601.14386; DOI: 10.1007/JHEP12(2024)216.
 - [10] KamLAND-Zen Collaboration (S. Abe *et al.*), “Search for Majorana neutrinos with the complete KamLAND-Zen dataset,” arXiv:2406.11438 (2024; v2 March 2026).
 - [11] LEGEND Collaboration (N. Abgrall *et al.*), “LEGEND-1000 Preconceptual Design Report,” arXiv:2107.11462 (2021).
 - [12] J. Tooby-Smith, “HepLean: digitalising high energy physics,” *Comput. Phys. Commun.* **308**, 109457 (2025); arXiv:2405.08863; DOI: 10.1016/j.cpc.2024.109457.
 - [13] S. Navas *et al.* (Particle Data Group), “Review of Particle Physics,” *Phys. Rev. D* **110**, 030001 (2024); DOI: 10.1103/PhysRevD.110.030001.

- [14] L. de Moura and S. Ullrich, “The Lean 4 theorem prover and programming language,” *Lect. Notes Comput. Sci.* **12699**, 625 (2021); DOI: 10.1007/978-3-030-79876-5_37.
- [15] The Mathlib Community, “The Lean Mathematical Library,” *Proc. CPP 2020*, 367 (2020); DOI: 10.1145/3372885.3373824.